

Supplementary Material for Local Evidence Is Not Completion: Evidence-Condition Geometry for Dynamic Agent Workflows

Anonymous Submission

Scope of the Supplement

This anonymous supplementary material contains reproducibility and audit details that are intentionally kept out of the main technical-content pages. These tables support the evidence-condition controller evaluation but are not used as additional main effectiveness evidence. In particular, the workflow pilot is a workflow-shape/interface validation only, and the scalar singleton/Chao proxy is a negative control because it does not encode source-route evidence conditions.

Table 1: Negative control: scalar singleton/Chao proxy on homogeneous runs. The proxy observes item-count sparsity but does not encode source-route condition, so it is not a competitive controller baseline for certificate mismatch.

Task	Observed	Oracle total	Recall	f_1	f_2	f_1/obs	Chao1	Scalar stop	False if stop
policy_docset_v1	17	24	0.708	17	0	1.000	153.0	No	No
code_repo_v1	6	20	0.300	6	0	1.000	21.0	No	No
requests	31	298	0.104	31	0	1.000	496.0	No	No
urllib3	135	699	0.193	135	0	1.000	9180.0	No	No

Table 2: External oracle construction summary. Pattern-defined external oracles are line-level source-route evidence occurrences from frozen local snapshots.

Repo	Version	Items	Routes	Nonempty strata	Granularity
requests	2.32.5	298	4	18	line-level source-route
urllib3	2.5.0	699	5	24	line-level source-route

Table 3: Small logged workflow pilot for workflow-shape/interface validation only. The pilot fixes the prompt and records independent contexts, action/evidence events, localized stop proposals, and the controller-facing decision record; it is not used as main effectiveness evidence.

Task	Model	Agents	Independent contexts	Action events	Evidence events	Stop proposals	Controller decision
T_doc_dynamic_workflow_smoke	logged LLM-agent workflow	3	3	39	35	3	not scored; workflow-shape validation

Table 4: Seeded safe-state validation. These are oracle-safe broad-complete states with seeded order perturbations; the controller should remain SAFE rather than over-reacting to extra repair opportunity.

Task	State	Order/budget sweep	Runs	#SAFE	#CONTINUE	#ABSTAIN	FCR	Safe cov.	Repair gain	Mean cost	Cost range
requests	route_partitioned	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3041.2	157-7647
urllib3	extended_audit	5 orders; budgets 1-8	3000	3000	0	0	0.000	1.000	0.0	3678.2	275-9072

Table 5: Route-match consistency sanity check for external oracle positives. A small sample of line-level positives was checked for route consistency; ambiguous lines are retained as such rather than treated as full validation.

Repo	Route	Sample	Positive	Precision	Ambiguous	Notes
requests	compat_route	8	8	1.000	0	line-level oracle-positive sample
requests	exception_route	8	8	1.000	0	line-level oracle-positive sample
requests	timeout_route	8	8	1.000	0	line-level oracle-positive sample
requests	tls_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	cleanup_route	8	7	0.875	1	line-level oracle-positive sample
urllib3	exception_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	retry_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	timeout_route	8	8	1.000	0	line-level oracle-positive sample
urllib3	tls_route	8	8	1.000	0	line-level oracle-positive sample

Table 6: Leakage control for controller and repair decisions. Runtime decisions use only visible evidence. Oracle labels, oracle totals, post-hoc recall, and undiscovered true item counts are used only after trajectories and challenger choices are fixed.

Component	Runtime allowed	Oracle forbidden
Exposure	visits, scans, actions	missing mass
Support/Gini	exposure counts	post-hoc recall
Potential	text, routes, matches	hidden true items
Decision	condition and repair	target distribution

Table 7: External oracle appendix detail. Each route uses a frozen regex pattern; items are deduplicated by source, route, and line. To avoid unreadable regex truncation, the appendix reports a short pattern category and one representative line-level item per route; full regex rules and items are in the generated CSVs.

Repo	Route	Items	Pattern category	Dedup	Representative item
requests	tls_route	102	TLS/certificate/verification terms	source+route+line	adapters:tls_route:103: # cert path
requests	timeout_route	31	timeout/connect/read-timeout terms	source+route+line	adapters:timeout_route:120: self, request, stream=False, timeout=None, verify=True, cert=None, proxies=None
requests	exception_route	121	exception/raise/error terms	source+route+line	adapters:exception_route:136: raise NotImplementedError
requests	compat_route	44	compatibility/deprecation terms	source+route+line	adapters:compat_route:32: from .compat import basestring, urlparse
urllib3	timeout_route	135	timeout/connect/read-timeout terms	source+route+line	connection:timeout_route:137: timeout: .TYPE.TIMEOUT = _DEFAULT_TIMEOUT,
urllib3	retry_route	168	retry/backoff/status terms	source+route+line	connectionpool:retry_route:1011: retries,
urllib3	tls_route	111	TLS/certificate/verification terms	source+route+line	connection:tls_route:1005: cert,
urllib3	exception_route	170	exception/raise/error terms	source+route+line	connection:exception_route:1015: except BaseException:
urllib3	cleanup_route	115	close/release/finally terms	source+route+line	connection:cleanup_route:1016: ssl_sock.close()